

CHESSICA

AprilTag-Based Chess Piece Recognition for Autonomous Human-Robot Chess Using a UR5e Collaborative Arm

Introduction

This project presents an autonomous chess-playing system where a UR5e robot arm competes against a human opponent, with a PLC bridging the software and hardware sides of the system.

Method

A USB camera tracks board state via AprilTag detection, feeding into Chessica, a custom interface running Stockfish for move decisions. Moves are sent to a PLC over OPC-UA, then forwarded to the UR5e via TCP for execution with a parallel jaw gripper. The main challenges were reliable tag detection and PLC-to-robot communication.

Result

The system achieves highly reliable piece detection and completes moves in roughly 4 seconds, resulting in a smooth and functional game against a human opponent.

